

High-Resolution Feature Pyramid Network for Small Object Detection on Drone View

Zhaodong Chen¹, Student Member, IEEE, Hongbing Ji¹, Senior Member, IEEE, Yongquan Zhang, Zhigang Zhu¹, Member, IEEE, and Yifan Li¹

Abstract—Object detection has developed rapidly with the help of deep learning technologies recent years. However, object detection on drone view remains challenging due to two main reasons: (1) It is difficult to detect small-scale objects lacking detailed information. (2) The diversity of camera angles of drones brings dramatic differences in object scale. Although feature pyramid network (FPN) alleviates the problem caused by scale difference to some extent, it also retains some worthless features, which wastes resources and slows down the speed. In this work, we propose a novel High-Resolution Feature Pyramid Network (HR-FPN) to improve the detection accuracy of small-scale objects and avoid feature redundancy. The key components of HR-FPN include a high-resolution feature alignment module (HRFA), a high-resolution feature fusion module (HRFF) and a multi-scale decoupled head (MSDH). HRFA feeds multi-scale features from backbone into parallel resampling channels to obtain high-resolution features at the same scale. HRFF establishes a bottom-up path to distribute context-rich low-level semantic information to all layers that are then aggregated into classification feature and localization feature. MSDH cope with the scale difference of objects by predicting the categories and locations corresponding to different scales of objects separately. Moreover, we train model by scale-weighted loss to focus more on small-scale objects. Extensive experiments and comprehensive evaluations demonstrate the effectiveness and advancement of our approach.

Index Terms—Object detection on drone view, small object detector, high-resolution feature, multiple-in-single-out feature pyramid network.

I. INTRODUCTION

ALONG with the popularity of Unmanned Aerial Vehicles (UAVs), object detection technology on drone-captured scenarios are playing an increasingly important role in many practical applications, such as ecological protection [1], agriculture production [2], public safety [3], traffic monitoring [4]. In this paper, we work on improving the accuracy and efficiency of small-scale object detection on drone view.

Manuscript received 24 May 2023; accepted 10 June 2023. Date of publication 16 June 2023; date of current version 8 January 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62203343 and Grant 62276204. This article was recommended by Associate Editor J. U. Kim. (Corresponding author: Hongbing Ji.)

The authors are with the Xi'an Key Laboratory of Intelligent Spectrum Sensing and Information Fusion, and the Shaanxi Union Research Center of University and Enterprise for Intelligent Spectrum Sensing and Information Fusion, Xidian University, Xi'an 710071, China (e-mail: zdchen@stu.xidian.edu.cn; hbji@xidian.edu.cn; zhangyq@xidian.edu.cn; zgzh@xidian.edu.cn; leon63963@163.com).

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/TCSVT.2023.3286896>.

Digital Object Identifier 10.1109/TCSVT.2023.3286896

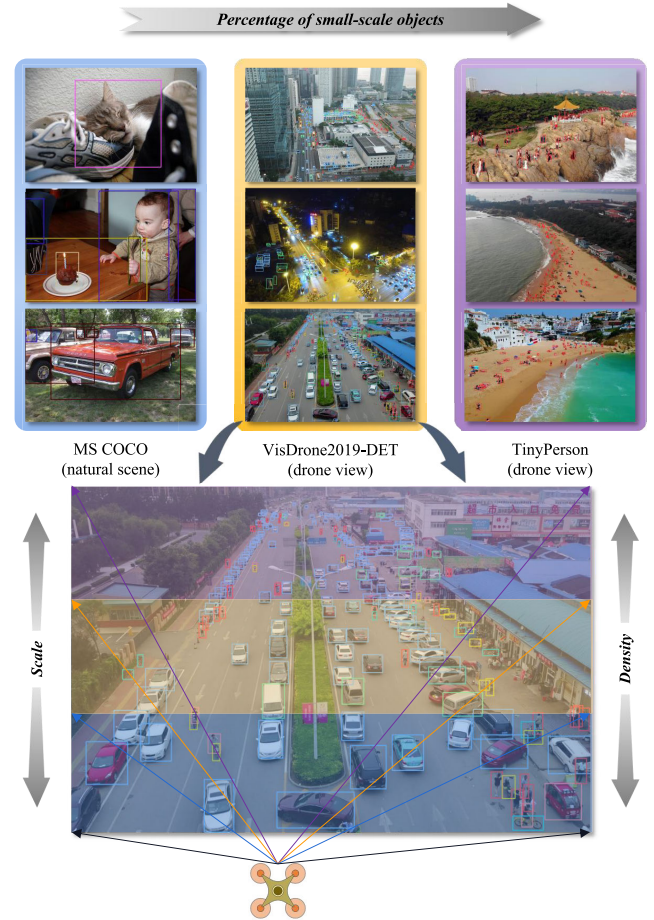


Fig. 1. The upper half shows the difference in the percentage of small-scale objects between the three datasets. Here, the area of the small-scale object is less than 32×32 pixel. The bottom half shows the vast differences in scale and density of objects in the scenes taken from the tilted perspective of the drone.

In the last decade, deep learning techniques have been rapidly gaining popularity in various fields [5], [6], [7], [8]. Now, object detection using deep neural networks has also become mainstream [9], [10], [11]. However, the direct application of these deep detectors on drone-captured images leads to a number of problems, as summarized below. First, these images contain more tiny objects with low resolution and less external features, which leads to poor performance of the general detectors. Fig. 1 shows the dramatic difference between the drone-captured dataset (TinyPerson [12] and

VisDrone2019-DET [13]) and the natural scene datasets (MS COCO [14] and Pascal VOC [15]) in terms of the percentage of small objects. Second, the diversity of camera angles of drones leads to huge differences in the scale and density of objects within the same scene, as shown in Fig. 1. What's more, in the pursuit of higher scores, advanced networks often consume numerous resources [16], [17], [18], which is far from the real-time detection required by drones. We facilitate small-scale object detection while avoiding unnecessary computations by optimizing the structure of feature pyramid network.

Among most deep detectors, feature pyramid network (FPN) [19], which are robust to multi-scale objects, has become a fundamental structure and been proven to be suitable for object detection in natural scenes. But FPN does not perform as well in small-scale object scenarios captured by drones. In this paper, we design a set of controlled variable experiments to investigate the performances of different FPN structures on the TinyPerson dataset. First, the FPN is considered as a multiple-input multiple-output architecture to build a simple model. On this basis, we then set up four FPN structures, namely Multiple-in-Multiple-out (MiMo), Multiple-in-Single-out (MiSo), Single-in-Multiple-out (SiMo), and Single-in-Single-out (SiSo), as shown in Fig. 2 (a) - (d). Interestingly, MiSo structure, which only has one output feature Y_2 and one prediction head, even outperforms MiMo structure a little, as shown in Fig. 2 (b). In contrast, the performance of SiMo and SiSo structures with only one input X_5 decreases sharply, as shown in Fig. 2 (c) and (d). It is worth noting that our experimental results are the opposite of those in [20]. This is because the TinyPerson dataset consists mainly of small-scale objects lacking detail information, which is quite different from the natural scene datasets such as MS COCO. These phenomenons suggest two facts: (1) Multi-scale feature fusion contributes a lot to small-scale object detection; (2) The high-level features take little contribution to detect small-scale objects.

According to the above findings, we construct the UP-MiSo structure by remodeling the low-resolution high-level features that are unprofitable for small-scale object detection, as shown in Fig. 2 (e). First, all features from backbone are upsampled to the same resolution as X_2 . Then, these high-resolution features are aggregated and fed directly into the prediction head. In the test set of TinyPerson dataset, UP-MiSo improves 1.4% than MiSo on AP , which illustrates the advantage of upsampling the features. More experimental results are shown in row 3 of Table II. Note that the MiSo structure discards most of the depth features, which causes information loss to some extent. To take better advantage of FPN, we construct UP-MiSo, which upsamples low-resolution features to the same size and then fuses them, which preserves information and increases the probability of finding small objects.

Continuing in this vein, we design a novel and effective High-Resolution Feature Pyramid Network (HR-FPN) to elevate the accuracy of small-scale object detection and avoid the redundancy of features. HR-FPN consists of three components: a high-resolution feature alignment module (HRFA), a high-resolution feature fusion module (HRFF) and a multi-scale

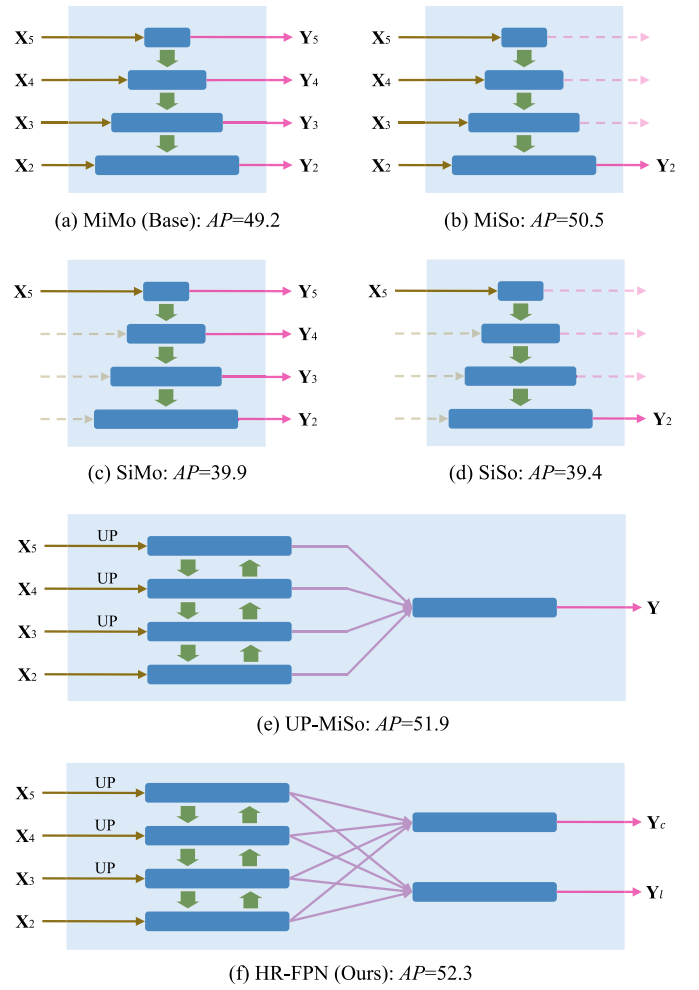


Fig. 2. Comparison of AP (IoU threshold of 0.5) scores on TinyPerson test set for several different structures of FPN. The downsample rates of X_2 , X_3 , X_4 , and X_5 are {4, 8, 16, 32}, respectively. The above experiments are uniformly based on YOLOv5 architecture whose backbone is CSP-DarkNet. All models are trained with the same settings. In the test phase, the confidence threshold is set to 0.25 and the IoU threshold of non-maximum suppression (NMS) is set to 0.5, the input images are resize to 1024×1024 . "UP" in (e) and (f) denotes the upsampling module containing a convolution block. The score of MiMo is regarded as baseline.

decoupled head (MSDH). First, HRFA resamples the feature maps from different levels to the same resolution as X_2 . During this process, top-down cross-connections are designed to enhance the information communication between the layers. A context extractor is used to extract both global and local contextual information of X_2 . Then, HRFF is responsible for fusing the aligned features. To avoid the conflict between classification task and localization task, these high-resolution features are aggregated into a classification feature Y_c and a localization feature Y_l . Although it helps a lot in detecting small-scale objects, this single-scale output structure is not conducive to resolving object scale differences. Here, MSDH predicts the categories and locations of objects at different scales by decoupling Y_c and Y_l .

Collaborating with CSP-DarkNet [21] (backbone), our HR-FPN achieves the best performance in TinyPerson test set among the above architectures and improves 3.1% than FPN on AP , as shown in Fig. 2 (f). Extensive experiments and

comprehensive evaluations also demonstrate the effectiveness and advancement of the proposed approach for small-scale object detection in drone's view. The contributions of our work are summarized as follows:

- By analyzing the limitations of traditional FPN for small object detection, we propose a novel structure, High-Resolution Feature Pyramid Network (HR-FPN), which can improve the probability for discovering small-scale objects and avoid the feature redundancy of vanilla FPN by improving and unifying the resolution of multi-scale features.
- We develop two novel and efficient components, high-resolution feature alignment (HRFA) and high-resolution feature fusion (HRFF), which are responsible for the alignment and fusion of high-resolution features, respectively.
- We design a multi-scale decoupled head (MSDH) to predict the categories and locations corresponding to different scales of objects separately, which can cope with the huge difference of anchor sizes due to the single-scale feature from HRFF.
- Based on the above modules, our HR-FPN reaches the state-of-the-art on TinyPerson dataset and VisDrone2019-DET dataset with fewer parameters and computing resources.

II. RELATED WORK

Most of the small object detectors are improved from the general ones. Among these methods, multi-scale feature fusion, decoupled prediction head and the use of context information attract more attention. Next, we focus on these related work in recent years.

A. Features Fusion In Small Object Detection

Currently, multi-scale feature fusion has been widely used in deep detectors. However, at the beginning of the deep learning revolution, CNN-based detectors employ only the last layer of feature from backbone to detect objects, such as Fast and Faster-RCNN [9]. Despite good results achieved on large-scale objects, they are not sufficient for the scenarios with long span of object scales because the single high-level feature contain less location information. Lin et al. [19] proposed feature pyramid network (FPN) to transfer high-level semantic information to each level features and exploit them to detect objects at different scales, which has achieved considerable success and drawn much attention. However, Liu et al. [22] concluded that the top-down fusion structure of FPN does not enrich the location information of high-level feature, and added a bottom-up branch, which improved the accuracy of object localization. Li et al. [23] proposed the cross-layer feature pyramid network (CFPN) to improve the progressive fusion effect of salient object detection by directly fusing the different layer features.

Although these methods extend the scale range of object detection, they still have unsatisfactory performance for small objects. Deng et al. [24] extended the FPN with a set of high-resolution features specifically for small object detection.

Leng et al. [25] designed a bidirectional feature fusion network to enhance the feature representation of small objects. What's more, Huang et al. [26] added multiple low-level feature branches on the FPN to fuse detail information. The essence of these improvements is to obtain more small object information by adding high-resolution feature maps, which can be computationally expensive. Considering this problem, Yang et al. [27] proposed QueryDet to balance accuracy and speed by accelerating high-resolution small object detection. In the FS-SSD [28], the two feature pyramids generated by the deconvolution module and feature fusion module are utilized to make predictions together, which can spike the model parameters and computational effort. The same problem occurs in [29], CADNet improves the recall of small objects by fusing deeper level feature maps from different channels. On the contrary, our proposed HR-FPN improves small object detection performance by optimizing the network structure and discarding redundant features, which does not increase the parameters.

B. Context Learning In Small Object Detection

It is shown that proper modeling of the context is beneficial for object detection [30]. Small objects lack obvious appearance features and require more context information to assist detection. Tang et al. [31] designed a new anchor that simultaneously extracts the objects and their surrounding features to improve the detection accuracy of tiny faces. To tackle the challenge of small object detection, Lim et al. [32] extracted some features from different layers as context to augment the information of small objects. For the same purpose, Leng et al. [25] first improved the quality of the proposal regions with context information, and then learn the pair-wise relationships between the proposal regions to generate the associated global information as a way to obtain better classification results. In addition, the method of capturing context by expanding the receptive field contributes to small object detection [33]. Nie et al. [34] extract multi-scale context features selectively with an enhanced context module and a triple attention module.

Vision Transformer (ViT) [35] holds a huge receptive field, which is a natural advantage in capturing contextual information [36]. But using self-attention alone results in missing location information. Fortunately, ACmix [37] made up for this deficiency by combining attention and convolution. Thus, we exploit ACmix module to capture the global and local context in high-resolution features to supplement location information for all layers.

C. Decoupled Head In Small Object Detection

Vanilla prediction heads predict objects' categories and locations with the same feature, which triggers conflict between classification task and regression task [38]. To mitigate this conflict, [10], [39] designed two sub-networks without sharing parameters to perform object classification and bounding box regression, respectively. Tian et al. [40] applied the decoupled prediction head to full convolution object detection

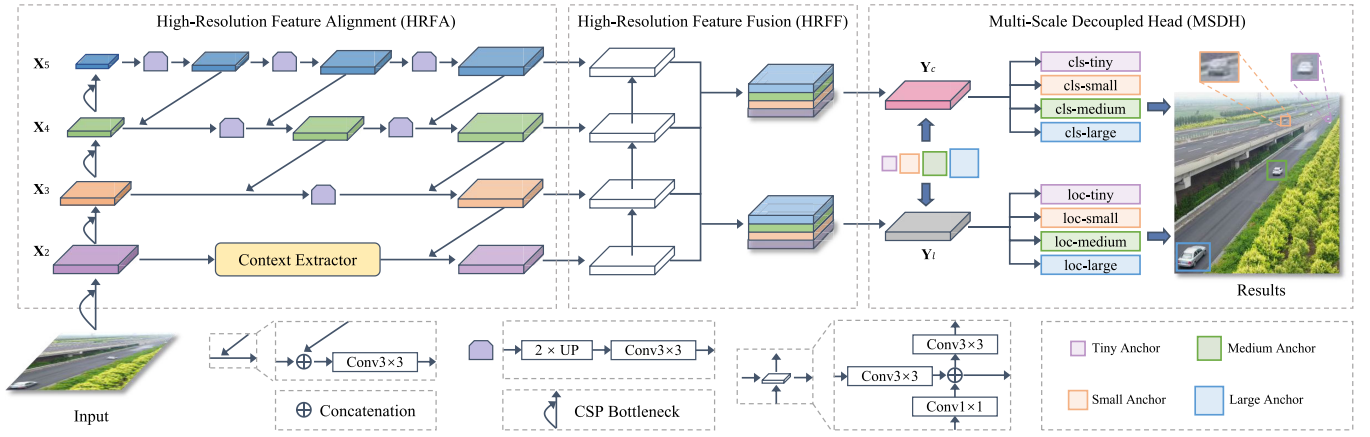


Fig. 3. Overall framework of our HR-FPN. The high-resolution feature alignment module (HRFA), high-resolution feature fusion module (HRFF) and multi-scale decoupled head (MSDH) are inserted into the vanilla feature pyramid network to improve the performance of small-scale object detection. Details of HRFA and HRFF are presented in Section III-B and Section III-C and MSDH are shown in Fig. 5 and Section III-D.

while Zhu et al. [41] used it for the anchor-free based detector. What's more, the decoupled head played an important role in small and dense object detection on remote sensing images [42].

However, our HR-FPN outputs high-resolution features at a single scale, which results in a prediction head that needs to cope with huge object scale differences. It is not sufficient to rely on the existing decoupled prediction heads. In this paper, our MSDH is designed to solve this difficulty.

III. OUR METHOD

In this section, we present our approaches to optimize small-scale object detection. First, section III-A gives an overview. Then HRFA and HRFF are described in detail in section III-B and section III-C. Next, section III-D introduces MSDH. finally, we introduce our method of training model by scale-weighted loss functions in section III-E.

A. Overview Architecture

Fig. 3 illustrates the overview architecture of High-Resolution Feature Pyramid Network (HR-FPN) that is composed of high-resolution feature alignment module (HRFA), high-resolution feature fusion module (HRFF) and multi-scale decoupled head (MSDH). First, HRFA gradually upsamples each layer feature maps from backbone to the same resolution as X_2 . There are top-down cross-connections between these up-sampling branches that permeates the high-level semantic information into the low-level features. In context extractor, we employ the hybrid module of attention and convolution (ACmix) [37] to extract the contextual information of X_2 in order to further mine small-scale objects. After completing feature alignment, HRFF aggregates these high-resolution features into two outputs, which are fed into the classification and localization sub-networks of MSDH, respectively. Finally, by decoupling between different scales, more objects are detected.

B. High-Resolution Feature Alignment

As mentioned earlier, the low-resolution high-level features output from vanilla FPN are weak in detecting small-scale

objects, thus reducing the performance of detector on TinyPerson test set, as illustrated in Fig. 2. However, EFPN [24] does not optimize the low-resolution high-level features when extending the pyramid layer, which limits the small-scale object detection performance. To improve the recall of small-scale objects, we upsample and align all the layers of features. Specifically, we propose a high-resolution feature alignment module (HRFA) that creates upsampling channels for each layer of raw features, gradually increasing the resolution to avoid mosaic effect. In this way, all the layers of features reach the same resolution, which provides a good foundation for feature fusion.

With CSP-DarkNet [21], [43] as backbone, we describe HRFA in detail. The resources increase with the improvement of feature resolution. Considering both computational consumption and model performance, we choose X_2 as the lowest-level feature. Denote the features output from the last four convolution blocks in backbone as $X_n \in \mathbb{R}^{C_n \times H_n \times W_n}$, $n \in \{2, 3, 4, 5\}$, C_n , H_n , and W_n represent the number of channels, height, and width of the n -th layer feature maps, respectively. They are set as

$$C_n = 2^{n+5}, H_n = H_0/2^n, W_n = W_0/2^n \quad (1)$$

where, H_0 and W_0 denote the height and width of the input image, respectively. The upsampling mode is nearest neighbor interpolation. Thus X_3 , X_4 , and X_5 need to be $2\times$ upsampled 1, 2, and 3 times, respectively. To avoid the mosaic effect from successive upsampling, we bridge a Residual block [44] after each interpolation. As the upsampling channels forward, top-down jump connections are interspersed between them to supplement high-level semantic information for low-level features. Formally, denote X_n^i as the intermediate variable of the n -th-layer feature in the upsampling channels, i denotes the number of upsampling. When $n = 5$, X_n^i is denoted as

$$X_5^i = \mathcal{R}(X_5^{i-1} \uparrow_{2\times}) \quad (2)$$

where $\mathcal{R}(\cdot)$ denotes the Residual block, $\uparrow_{2\times}$ denotes double upscaling by nearest interpolation, $i \in \{1, 2, 3\}$. When

$n \in \{3, 4\}$, $\mathbf{X}_n^i$ is denoted as

$$\mathbf{X}_n^i = \mathcal{R}(\mathbf{X}_n^{i-1} \| \mathcal{R}(\mathbf{X}_{n+1}^{i-1} \uparrow_{2\times}) \uparrow_{2\times}) \quad (3)$$

where $\|$ denotes the concatenation function, $1 \leq i \leq n-2$, $i \in \mathbb{N}$. Let $\mathbf{P}_n \in \mathbb{R}^{C_n \times H_2 \times W_2}$ denotes the output feature of HRFA, and the formula is expressed as follows

$$\mathbf{P}_n = \mathcal{C}(\mathbf{X}_n^{n-2}) \quad (4)$$

where $\mathcal{C}(\cdot)$ denotes 1×1 convolution layer containing a SiLU activation function, $n \in \{3, 4, 5\}$. For $\mathbf{X}_2$, it retains abundant low-level semantic information of objects due to undergoing few convolution layers. To take full advantage of these original features, a context extractor is designed to extract global and local contextual information in $\mathbf{X}_2$. The context extractor consists of a ACmix module [37] and a 1×1 convolution layer, as shown in Fig. 4. The multi-head self attention branch and convolution branch in ACmix are used to extract global and local contextual information, respectively. The output feature $\mathbf{P}_2$ can be denoted as

$$\mathbf{P}_2 = \mathcal{C}(\mathcal{A}(\mathbf{X}_2) \| \mathbf{X}_3^1) \quad (5)$$

where $\mathcal{A}(\cdot)$ denotes the ACmix module which consists of a shifted window transformer block [45] and a 3×3 convolution layer.

Driven by horizontal upsampling channels and vertical jump connections, HRFA finally generates aligned high-resolution features $\{\mathbf{P}_2, \mathbf{P}_3, \mathbf{P}_4, \mathbf{P}_5\}$ preparing for fusion.

C. High-Resolution Feature Fusion

Given the high-resolution features $\mathbf{P}_n$ from HRFA, the easiest fusion method is to concatenate them together directly, as Fig. 2 (e). However, the high-level feature maps are bound to be blurred to some extent after multiple upsampling, which can affect the localization accuracy. In addition, this single output approach is unable to resolve the conflict between classification and localization tasks.

Therefore, we design a simple and effective module named High-Resolution Feature Fusion (HRFF), as shown in Fig. 3. First, a bottom-up path is established in order to transfer low-level semantic information to high-level features. Formally, the new born feature $\mathbf{P}'$ can be expressed as

$$\mathbf{P}'_n = \mathcal{R}(\mathbf{P}_{n-1} \| \mathbf{P}_n) \quad (6)$$

where $n \in \{3, 4, 5\}$. Then, we employ two 3×3 convolutions to map features into a set of classification feature $\mathbf{Y}_c$ and a set of localization feature $\mathbf{Y}_l$, expressed as

$$\mathbf{Y}_c = \mathcal{M}_c(\mathbf{P}_2 \| \mathbf{P}'_3 \| \mathbf{P}'_4 \| \mathbf{P}'_5) \quad (7)$$

$$\mathbf{Y}_l = \mathcal{M}_l(\mathbf{P}_2 \| \mathbf{P}'_3 \| \mathbf{P}'_4 \| \mathbf{P}'_5) \quad (8)$$

where $\mathcal{M}_c(\cdot)$ and $\mathcal{M}_l(\cdot)$ denote the classification feature mapper and the location feature mapper, respectively. We experimentally verify that this separation mapping can improve the efficiency of decoupled prediction head, as shown in Table II.

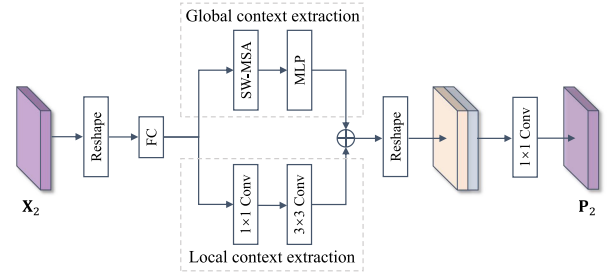


Fig. 4. Detailed illustration of context extractor. “FC” denotes the fully connected layer; “SW-MSA” denotes the shifted windows multi-head self attention; “MLP” denotes the multi-layer perceptron.

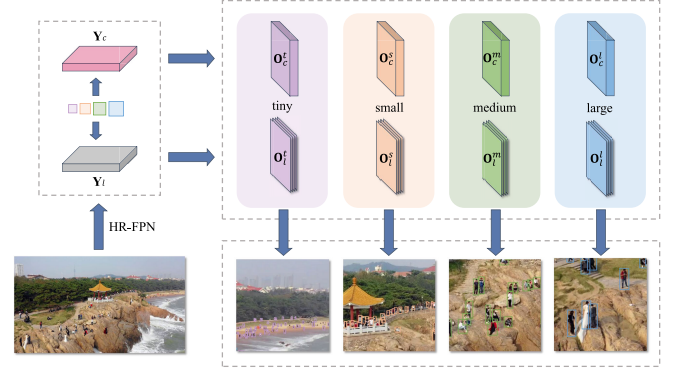


Fig. 5. Detailed illustration of the proposed multi-scale decoupled head (MSDH). $\mathbf{Y}_c$ and $\mathbf{Y}_l$ denote classification feature and localization feature, respectively. Objects are separated into four sets regarding their scales (tiny, small, medium, large).

D. Multi-Scale Decoupled Head

Conventional FPN deal with the scale difference of objects using “divide-and-conquer” [20], namely detecting objects on different levels regarding the scale of objects. Then, there are special prediction head for each scale of objects to predict the categories and locations. In general, the anchors within the same prediction head are of similar scales. However, our proposed HR-FPN only outputs high-resolution features at the same scale, which leads to the scales of anchors within the corresponding prediction head vary widely. Therefore, vanilla prediction head can’t cope with the scale differences in terms of HR-FPN.

We propose a simple and novel Multi-Scale Decoupled Head (MSDH) to solve the above problem. As shown in Fig. 5, MSDH contains four different scales of anchors (tiny, small, medium and large). First, the categories and locations of objects at different scales are decoupled from $\mathbf{Y}_c$ and $\mathbf{Y}_l$ by different scales of anchors:

$$\mathbf{O}_c^t = \mathcal{D}_c^t(\mathbf{Y}_c), \mathbf{O}_l^t = \mathcal{D}_l^t(\mathbf{Y}_l) \quad (9)$$

$$\mathbf{O}_c^s = \mathcal{D}_c^s(\mathbf{Y}_c), \mathbf{O}_l^s = \mathcal{D}_l^s(\mathbf{Y}_l) \quad (10)$$

$$\mathbf{O}_c^m = \mathcal{D}_c^m(\mathbf{Y}_c), \mathbf{O}_l^m = \mathcal{D}_l^m(\mathbf{Y}_l) \quad (11)$$

$$\mathbf{O}_c^l = \mathcal{D}_c^l(\mathbf{Y}_c), \mathbf{O}_l^l = \mathcal{D}_l^l(\mathbf{Y}_l) \quad (12)$$

where $\mathbf{O}_c^t, \mathbf{O}_c^s, \mathbf{O}_c^m$, and $\mathbf{O}_c^l$ denote the category sets of objects at four scales, and $\mathbf{O}_l^t, \mathbf{O}_l^s, \mathbf{O}_l^m$, and $\mathbf{O}_l^l$ denote the location sets of objects at four scales, respectively. $\mathcal{D}_c^t, \mathcal{D}_c^s, \mathcal{D}_c^m, \mathcal{D}_c^l$ and $\mathcal{D}_l^t, \mathcal{D}_l^s, \mathcal{D}_l^m, \mathcal{D}_l^l$ denote the decoupling operations. Then, MSDH

TABLE I
PERCENTAGE OF OBJECTS AT EACH SCALE IN DIFFERENT DATASETS.
FOR CONSISTENCY, ALL IMAGES ARE UNIFORMLY RESIZED TO
1024 × 1024. “TINY” DENOTES OBJECTS SMALLER THAN 16 × 16 IN
SIZE. “SMALL” DENOTES OBJECTS BETWEEN 16 × 16 AND 32 × 32 IN
SIZE. “MEDIUM” DENOTES OBJECTS BETWEEN 32 × 32 AND 64 × 64
IN SIZE. “LARGE” DENOTES OBJECTS LARGER THAN 64 × 64 IN SIZE

Datasets	Tiny	Small	Medium	Large
General Datasets:				
MS COCO [14]	3.1 %	9.7 %	19.1 %	68.1 %
Pascal VOC [15]	0.1 %	0.9 %	5.8 %	93.2 %
OID [47]	2.5 %	7.8 %	15.8 %	73.9 %
Drone View Datasets:				
TinyPerson [12]	80.7 %	15.0 %	3.5 %	0.8 %
UAVDT [48]	3.0 %	37.1 %	46.5 %	13.3 %
VisDrone2019-DET [13]	32.7 %	35.6 %	22.6 %	9.1 %

merges the prediction sets at different scales and outputs the final results after Non-Maximum Suppression (NMS).

E. Scale-Weighted Loss

From Table I, we can see that small-scale objects dominate in the scenes captured by drones. To make HR-FPN better focus on small-scale objects, we use scale-weighted loss to train our model. The main idea of this loss is to multiply different weights to the loss components of different scale samples in order to adjust the model to focus on objects at a specific scale. The scale-weighted loss is described below.

Reviewing III-B, MSDH contains four kinds of anchors, which are used to detect objects at four scales of tiny, small, medium, and large, respectively. In the training phase, for the input image $\mathcal{I}$, the bounding box loss of MSDH output can be expressed as

$$L_{box}^n = \frac{\sum_{p_i \in \mathcal{P}_l^n, y_i \in \mathcal{Y}_l} (1 - CIoU(p_i, y_i))}{N_b} \quad (13)$$

where, N_b denotes the number of bounding boxes in the input image. $\mathcal{P}_l^n \in \mathbb{R}^{N_b \times 4}$ denotes the set of prediction bounding boxes detected with anchors of different scales, and $n \in \{1, 2, 3, 4\}$. $\mathcal{Y}_l \in \mathbb{R}^{N_b \times 4}$ denotes the bounding box annotations. $CIoU(\cdot)$ denotes the Complete-IoU (CIoU) loss and its details can be found in [46]. Next, we weight all the bounding box loss components and then sum them to obtain the final bounding box loss:

$$L_{box} = \frac{\sum_{n=1}^{N_a} w_n \cdot L_{box}^n}{N_a} \quad (14)$$

where N_a denotes the number of anchor scales, which is set to four. w_n denotes the scale weight. We aim to make the model focus on small-scale objects by adjusting the loss components of objects at different scales. Therefore, w_n should be positively correlated with the proportion of objects at corresponding scale in the dataset. We design the formula to

calculate w_n as

$$w_n = 1 + \frac{2}{\pi} \arctan(\ln(r_n \cdot N_a)) \quad (15)$$

where, $r_n \in [0, 1]$ denotes the percentage of objects at the n -th scale, $r_n = N_n/N_b$, and N_n denotes the number of n -th scale annotation boxes. The value of N_n is obtained by clustering the area of the labeled objects in the dataset. Compared with the direct representation of w_n by r_n , the nonlinear mapping of Eq. (15) avoids too much difference in w_n values, which ensures the stability of training.

Similarly, the classification loss section of scale-weighted loss is formulated as

$$L_{cls} = \frac{\sum_{n=1}^{N_a} w_n \cdot L_{bce}(\mathcal{P}_c^n, \mathcal{Y}_c)}{N_a} \quad (16)$$

where, $\mathcal{P}_c^n \in \mathbb{N}^{N_b \times 1}$ denotes the set of predicted object categories by anchors of different scales. $\mathcal{Y}_c \in \mathbb{N}^{N_b \times 1}$ denotes the category labels. $L_{bce}(\cdot)$ denotes the binary cross entropy (BCE) loss function.

Ultimately, the complete scale-weighted loss is expressed as

$$L = \lambda_b \cdot L_{box} + \lambda_c \cdot L_{cls} + \lambda_o \cdot L_{obj} \quad (17)$$

where, L_{obj} denotes the confidence loss, which is also calculated by the BCE loss function. λ_b , λ_c and λ_o are three decay coefficients.

IV. EXPERIMENTS

In this section, we first give the detail experimental configurations, including datasets, hyperparameter setting and evaluation metrics. Then, we evaluate the contribution of each component separately by ablation study. Next, we test our model on several drone-captured datasets and compare it with some advanced models. At last, the efficiency of our network is evaluated in term of parameters and computational cost.

A. Datasets

1) *TinyPerson*: This is the first benchmark for person detection at a distance and in a large context, opening up a new promising direction for tiny object detection. The dataset consists of 1610 images containing more than 200 persons each, with the train set containing 794 images and the test set containing 816 images. TinyPerson has 72651 manually labeled tiny persons, which are classified into five categories: “sea”, “earth”, “ignore”, “uncertain”, and “dense”. For convenience, we regard all labeled tiny persons as a “person” category. The test set is used for evaluation.

2) *UAVDT*: The UAVDT dataset [48] contains 40735 drone-view images with a size of 1024 × 540. There are three categories in labeled objects: “car”, “bus” and “truck”. We randomly split the dataset into train set consists of 24441 images and test set consists of 16294 images. The UAVDT dataset is mainly used to evaluate the performances of models in ablation studies.

TABLE II

RESULTS OF ABLATION STUDIES ON THE TEST SET OF TINYPERSON. “✓” INDICATES THAT THE COMPONENT IS EMPLOYED. THE SIZE OF INPUT IMAGE IS 1024×1024 . WEIGHTED LOSS INCLUDES WEIGHTED BINARY CROSS ENTROPY LOSS AND WEIGHTED CIOU LOSS. THE “SW LOSS” DENOTES SCALE-WEIGHTED LOSS. THE “PARAMS” DENOTES THE NUMBER OF MODEL PARAMETERS. BOLD FONT INDICATES THE BEST RESULT OF ONE METRIC. THE BASELINE IS YOLOV5L

Method	HRFA	HRFF	MSDH	SW Loss	AP^{t1}	AP^{t2}	AP^t	AP^{s1}	AP^{s2}	AP^s	AP^m	AP^l	AP	Params
Baseline					25.1	41.2	36.2	51.9	57.5	53.6	66.0	67.0	45.5	46.1 M
	✓				27.4	47.3	42.1	56.2	59.4	57.9	66.6	56.2	50.3	31.6 M
	✓	✓			27.9	49.2	43.5	57.7	62.3	59.2	68.3	58.6	51.9	32.0 M
	✓	✓	✓		27.8	49.9	43.7	58.9	64.7	60.7	70.2	60.3	52.3	32.1 M
	✓	✓	✓	✓	28.4	50.1	43.9	59.6	64.8	61.2	70.9	59.4	52.5	32.1 M

3) *VisDrone2019-DET*: This is a large dataset taken by drones flying over 14 cities in China, with a focus on small objects. The training set, validation set, and test set contain 6471, 548, and 1610 images, respectively. The 2.6 million object instances in VisDrone2019-DET are divided into 10 categories. We use test-dev set to evaluate our models.

B. Experiment Settings

1) *Implementation Details*: All our models are trained and tested on machines with 12-core processor, 128G RAM, a RTX 3090 GPU and pytorch 1.8, CUDA 11.2. In the training phase, all our models are trained on each dataset by SGD optimizer with batch size 4. In order to better balance localization and classification loss, we set λ_b , λ_c and λ_o in Eq. (17) to 0.05, 0.5 and 1.0, respectively. The learning rate updates from 0.01 by cosine decay strategy. There are large structural differences between HR-FPN and baseline (YOLOv5l), so we abandon the pretrained models. To highlight the advantages of proposed method on small-scale object detection, we train our network mainly on TinyPerson and VisDrone2019-DET, and compare it with other advanced detectors. For variable consistency, we use the same data augmentation operations as YOLOv5.¹ What’s more, some methods employ multi-scale train strategy with input resolution of $\{(1333, 640), (1333, 720), (1333, 800)\}$.

During the test phase, the confidence threshold is set to 0.25 and the IoU threshold is set to 0.5 for NMS. All tests are performed without test-time augmentations and other tricks. The input images are uniformly resized to 1024×1024 resolution in both phases. Our source code can be found at <https://github.com/laochen330/HR-FPN>.

2) *Evaluation Metrics*: Average precision (AP) and mean average precision (mAP) are used as evaluation metrics. We evaluate models by AP on TinyPerson dataset containing only one category while mAP on VisDrone2019-DET and UAVDT datasets containing multi categories. Note that the protocol in [14] only provides AP^s ($area < 32^2$) to measure small-scale object detection accuracy, which is clearly not enough. Unlike MS COCO, the drone-captured datasets generally contain a large number of objects smaller than 16^2 . Therefore, there should be special metrics for measuring the detection performance of each scale objects (tiny, small,

medium, large). Accordingly, we reformulate more precise evaluation metrics, namely AP^t ($area < 16^2$), AP^s ($16^2 < area < 32^2$), AP^m ($32^2 < area < 96^2$) and AP^l ($area > 96^2$). Further, AP^t is divided into AP^{t1} ($area < 8^2$) and AP^{t2} ($8^2 < area < 16^2$), also AP^s is divided into AP^{s1} and AP^{s2} . Here, the scale attributes of all objects are divided after the images are resized to 1024×1024 . Conveniently, the IoU threshold for the reported AP is uniformly set to 0.5 and the maximum detection per image is set to 1000 by default to avoid truncation of dense scenes.

C. Ablation Studies

Using YOLOv5l as baseline, ablation studies are performed to explore the effectiveness of each component, including HRFA, HRFF, and MSDH. In addition, we also investigate the impact of losses, backbones, upsampling modes and input sizes on network performance.

1) *Effectiveness of HRFA*: To control the variables, we simply replace the PANet in the original YOLOv5l with the HRFA module and directly aggregate the alignment features output from the HRFA. As can be seen in the row 2 of Table II, this component helps baseline gain 4.8% improvement on AP, 5.9% and 4.3% on AP^t and AP^s , respectively. The largest scale feature output from original YOLOv5l is X_3 while HRFA upsamples all features to the same resolution as X_2 , which is the key to improving the accuracy of small object detection. Although this method of increasing the feature resolution leads to a degradation of large object detection performance for the model, the overall detection performance is greatly improved. In addition, it is obvious from Table II that replacing the PANet in the original YOLOv5l with the HRFA module leads to a significant reduction in the model parameters, which is due to the fact that we use convolutions with few layers and few channels in our modules. This result also shows that deeper and wider networks are redundant for detecting small objects that lack detailed information. What’s more, compared with the PANet of Baseline, HRFA use the convolution layers with less channels, which makes the parameters of HR-FPN less. We also explore the effect of context extractor in HRFA. As shown in Table III, the addition of context extractor brings some improvements on most metrics for the model.

2) *Effectiveness of HRFF*: Following the above experiments, we evaluate the effectiveness of HRFF. Without MSDH, we concatenate the classification and localization features

¹<https://github.com/ultralytics/yolov5>

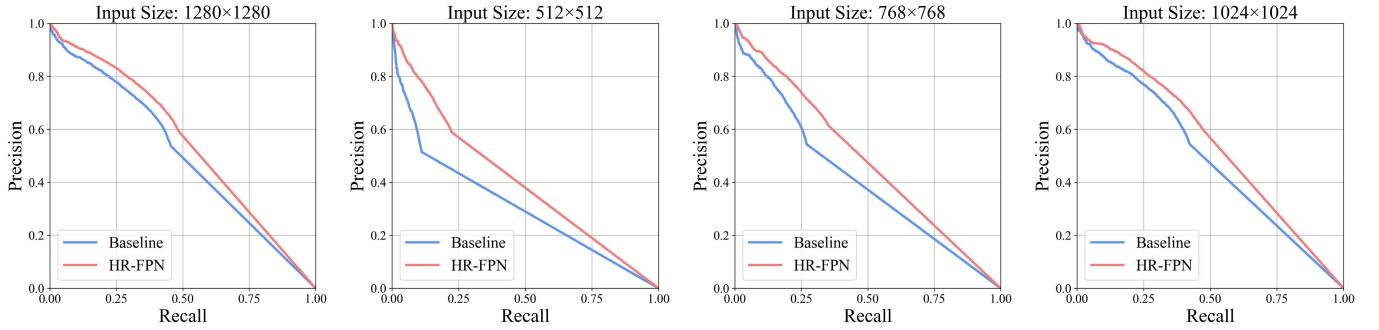


Fig. 6. Comparison of HR-FPN and YOLOv5 on TinyPerson test set at different input image sizes. The area under the Recall-Precision curve reflects the accuracy of the model.

TABLE III
COMPARISON RESULTS OF HR-FPN WITHOUT CONTEXT EXTRACTOR
AND THE FULL HR-FPN ON THE TEST SET OF UAVDT. THE
“CE” DENOTES CONTEXT EXTRACTOR

Methods	mAP^t	mAP^s	mAP^m	mAP^l	mAP
without CE	25.7	45.4	59.5	52.7	48.3
HR-FPN	26.1	45.3	59.9	54.1	48.5

output from HRFF together and feed them into the original prediction head. As shown in row 3 of Table II, with the contribution of HRFF, the model improves by another 1.4% and 1.3% on AP^t and AP^s , and by another 1.7% on AP^m , and 2.4% on AP^l . The greater improvement in the large object detection accuracy is mainly due to the fact that the bottom-up path propagates low-level semantic information to high-level features, which enhances the localization performance of the model for large objects.

3) *Effectiveness of MSDH*: Next, we evaluate the impact of MSDH on the results. From row 4 of Table II, the addition of MSDH barely improves small object detection accuracy, boosting 0.2% on AP^t , 0.5% on AP^s for compared to the results in row 3 while has more improvement in large objects, 1.9% higher on AP^m , and 1.7% higher on AP^l . In essence, multi-scale decoupling is to use the corresponding features of each scale anchor to predict the categories and locations of corresponding scale objects, which can well help HR-FPN deal with the difference of object scales.

4) *Effectiveness of Scale-Weighted Loss*: Then, we evaluate the contribution of scale-weighted loss function. As can be seen from row 5 of Table II, training HR-FPN with scale-weighted CIOU loss and scale-weighted BCE loss can improve the performance of small-scale object detection by a small margin. Our scale weights are obtained by clustering the bounding box annotations of train set, so they can adjust adaptively for any dataset.

5) *Effectiveness of Backbone*: We explore the effects of different backbones on HR-FPN, including ResNet-50, CSP-DarkNet, GhostNet [83] and CSP-DarkNet with Transformer block (ViT). As can be seen from Table IV, backbones have little effect on the final results, such as the largest difference in AP scores between HR-FPN using GhostNet and CSP-DarkNet, which is only 2.9% on AP . However, HR-FPN

improves 6.8% on AP compare to baseline, which indicates that it is our optimizations that play the main roles. Considering the model parameters, computational consumption and training time using different backbones, as Table IV, we choose HR-FPN with CSP-DarkNet as backbone to compare with other advanced models in the subsequent experiments.

6) *Effectiveness of Upsampling Mode*: We further explore the effect of upsampling mode in HRFA on HR-FPN, including Transpose-Convolution, Bilinear, Bicubic and Nearest. As shown in Table V, HR-FPN with Bicubic achieves the highest score on AP^t . With the help of nearest neighbor interpolation, HR-FPN get the highest scores on AP^s , AP^l and AP , and has the fastest inference speed. Incredibly, T-Conv hardly help HR-FPN improve the detection performances, probably because of the conflict between transpose convolution and the subsequent 3×3 convolution. All things considered, we employ nearest neighbor interpolation as the upsampling mode in HRFA.

7) *Effectiveness of Input Size*: We train and test the model using different input sizes as a way to explore the effect of input size on the results, as shown in Fig. 6. With input sizes from 512×512 to 768×768 , the accuracy of HR-FPN increases significantly, but from 1024×1024 to 1280×1280 , the improvements of performance hit a bottleneck. On balance, we choose 1024×1024 as the input size for subsequent experiments. What's more, the Precision-Recall (PR) curves of baseline and HR-FPN are getting closer with the increasing of input size, which indicates that baseline is more sensitive to input size changes compared to HR-FPN.

D. Comparison With State-of-the-Arts

We compare the proposed method with some advanced detectors. All models are trained with the same configurations as described in Section IV-B, and other details follow the codes provided by original authors.

1) *Qualitative Comparison*: The qualitative comparison results on the test set of VisDrone2019-DET and TinyPerson are shown in Fig. 7 and Fig. 8, respectively. Baseline is YOLOv5l and the FPN [19] is based on Faster R-CNN [9] with ResNet-50.

In Fig. 7, it is obvious that HR-FPN can accurately detect more tiny objects while dealing with scale changes, which

TABLE IV

RESULTS OF HR-FPN WITH DIFFERENT BACKBONES ON TINYPERSON TEST SET. ALL MODELS ARE TRAINED WITH THE SAME LOSS FUNCTIONS AND SETTINGS. HIGHER SCORES OF AP REPRESENT BETTER PERFORMANCE WHILE LOWER PARAMETERS AND GFLOPS REPRESENT LESS COMPUTATIONAL CONSUMPTION. THE VALUE OF “TRAIN TIME” REPRESENTS THE TIME CONSUMED PER TRAINING EPOCH. BOLD FONT INDICATES THE BEST RESULT OF ONE METRIC

Methods	Backbone	AP^{t1}	AP^{t2}	AP^t	AP^{s1}	AP^{s2}	AP^s	AP^m	AP^l	AP	Parameters	GFLOPs	Train-Time
HR-FPN	GhostNet	28.3	46.8	41.1	55.9	61.8	57.8	69.2	59.5	49.4	14.9 M	279.2	244.3 s
	ViT	27.1	48.1	42.5	57.4	62.2	58.9	71.8	62.1	50.8	31.6 M	396.5	89.1 s
	ResNet-50	28.2	49.0	43.0	57.0	64.2	59.3	68.5	59.3	51.2	39.6 M	443.1	87.5 s
	CSP-DarkNet	27.8	49.9	43.7	58.9	64.7	60.7	70.2	60.3	52.3	32.1 M	404.0	88.1 s

TABLE V

RESULTS OF HR-FPN WITH DIFFERENT MODES OF UPSAMPLE ON UAVDT TEST SET. ALL MODELS ARE TRAINED WITH THE SAME LOSS FUNCTIONS AND SETTINGS. HIGHER SCORE OF mAP REPRESENTS BETTER PERFORMANCE, LOWER GFLOPS REPRESENT LESS COMPUTATIONAL CONSUMPTION, AND HIGHER FPS REPRESENTS HIGHER INFERENCE SPEED. THE “T-CONV” DENOTES TRANSPOSE CONVOLUTION WITH 2×2 KERNEL. ALL MODES ARE $2 \times$ UPSAMPLING

Methods	Upsample mode	mAP^{t1}	mAP^{t2}	mAP^t	mAP^{s1}	mAP^{s2}	mAP^s	mAP^m	mAP^l	mAP	GFLOPs	FPS
HR-FPN	T-Conv	21.2	24.4	25.1	35.2	48.4	44.6	59.1	47.1	47.8	606.7	21.3
	Bilinear	20.4	25.4	25.6	34.9	48.8	44.5	60.2	52.7	48.1	401.6	23.3
	Bicubic	19.8	26.7	26.3	35.6	48.0	44.7	59.0	50.4	48.3	436.9	22.8
	Nearest	18.1	26.3	26.1	35.9	49.3	45.3	59.9	54.1	48.5	404.0	23.9

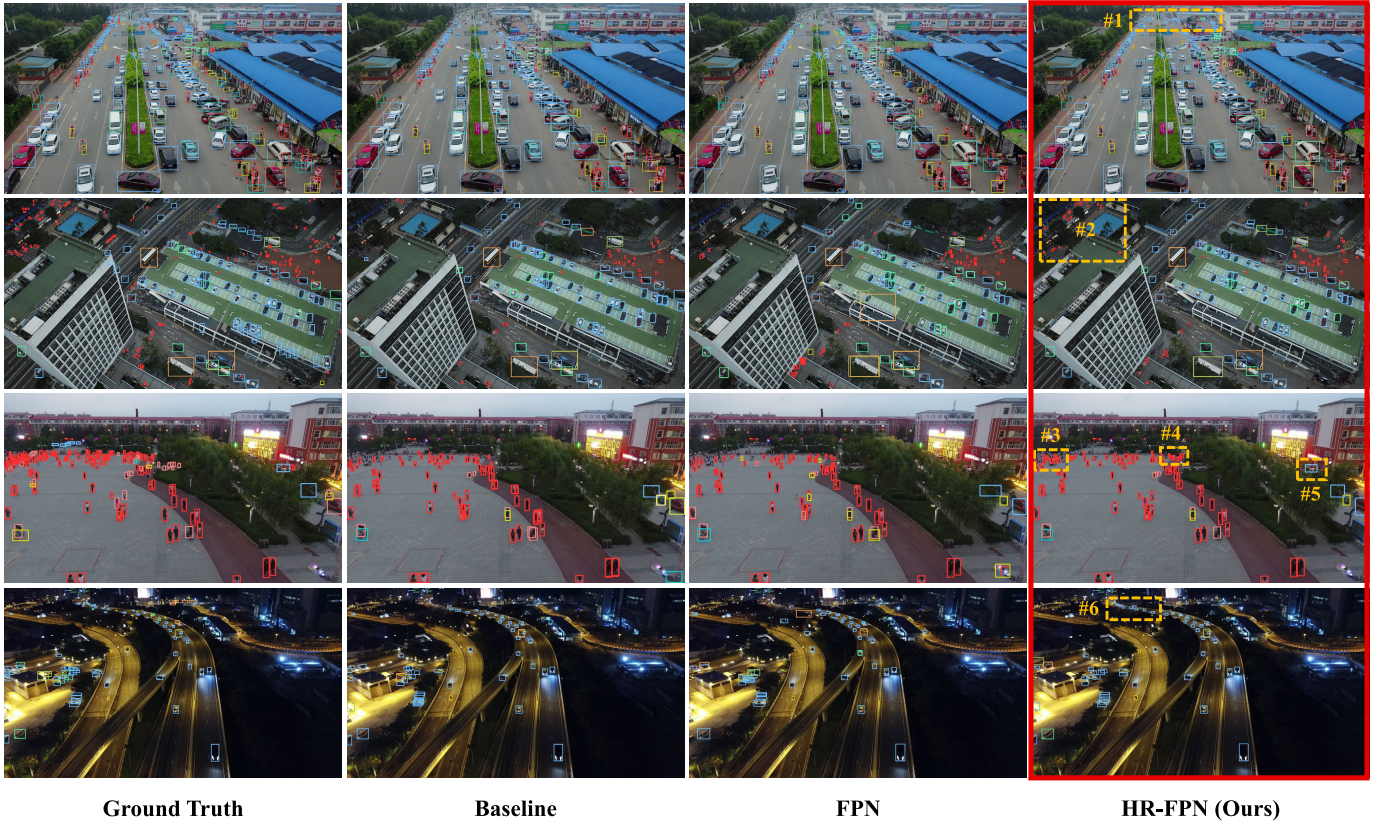


Fig. 7. The qualitative comparison results of different methods on the test set of VisDrone2019-DET dataset. For each testing image, we show the results of ground truth, baseline (YOLOv5l), FPN, and our HR-FPN. The orange dotted boxes show our superior performance for small-scale object detection in different scenarios.

expands the detection range when the camera is taking photographs at a tilted angle, as shown in regions #1 and #2. And our model is better at detecting tiny and dense targets, such

as the situations in regions #3 and #4. HR-FPN also detect a car heavily obscured by a tree while it is not found out by the other two models, as shown in region #5. Further, the

TABLE VI

QUANTITATIVE COMPARISON RESULTS OF PREVIOUS ADVANCED DETECTORS WITH OUR METHOD ON THE VALIDATION SET OF VisDrone2019-DET. “PEDESTRIAN” AND “PEOPLE” DENOTE THE AP SCORES OF “PEDESTRIAN” CATEGORY AND “PEOPLE” CATEGORY UNDER 0.5 IOU THRESHOLD. THE mAP DENOTES THE MEAN AP OF ALL CATEGORIES. BOLD FONT INDICATES THE BEST RESULT OF ONE METRIC

Methods	Backbone	Train Scale	Pedestrian	People	mAP^t	mAP^s	mAP^m	mAP^l	mAP
Two Stage Detectors:									
Dynamic R-CNN [49]	ResNet-50	Single	36.4	27.5	22.5	32.5	46.7	52.5	39.9
Libra R-CNN [50]	ResNeXt-101	Single	47.2	31.3	26.5	39.2	57.2	68.8	44.0
Faster R-CNN (FPN) [19]	RegNetX [51]	Single	49.1	34.5	27.9	40.6	56.1	64.0	44.7
Qiao <i>et al.</i> [52]	ResNeXt-101	Multi	49.8	35.2	32.2	39.2	57.8	65.3	45.1
PISA [53]	ResNeXt-101	Single	50.2	38.4	27.6	45.4	56.5	62.4	46.4
GRoIE [54]	ResNet-50	Multi	50.1	37.1	30.3	44.4	57.9	63.4	46.9
Cascade R-CNN [16]	HRNetV2p-W40 [18]	Multi	54.7	42.3	31.0	45.6	59.4	63.4	49.3
One Stage Detectors:									
CenterNet [55]	ResNet-18	Single	35.1	21.1	14.9	30.5	45.9	52.1	32.6
YOLOF [20]	ResNet-50	Single	30.9	19.3	20.2	27.1	44.0	58.3	33.2
Conditional DETR [56]	ResNet-50	Multi	31.5	18.9	15.7	27.0	44.7	60.8	33.7
AutoAssign [57]	ResNet-50	Single	36.8	21.0	16.4	32.9	50.4	59.1	35.4
DAB DETR [58]	ResNet-50	Multi	38.5	20.9	17.4	32.8	45.9	62.8	36.3
NAS-FCOS [59]	ResNet-50	Single	40.4	26.4	19.2	34.1	49.3	58.9	37.8
FreeAnchor [60]	ResNeXt-101	Single	38.6	25.2	20.5	38.3	52.7	62.8	39.5
Deformable DETR [61]	ResNet-50	Multi	39.3	24.7	21.7	39.7	54.0	63.6	41.1
TOOD [62]	ResNet-101	Multi	44.0	28.7	24.9	40.3	56.0	67.0	42.9
DINO [63]	Swin-L [45]	Multi	48.7	35.6	24.5	47.3	62.7	70.4	48.7
YOLOv5x	CSP-DarkNet	Single	51.6	37.9	32.1	46.7	62.9	62.9	49.1
VarifocalNet [64]	ResNet-101	Multi	51.2	38.3	30.8	45.8	62.0	70.2	49.8
HR-FPN (Ours)	CSP-DarkNet	Single	56.8	44.1	34.2	49.1	61.7	63.3	50.8

results of region #6 demonstrate that HR-FPN still has strong detection capability in night scenes with low contrast. These phenomenons reveal the strong robustness of HR-FPN.

The smaller object scale in TinyPerson leads to more failed detections by the previous advanced models. For example, baseline and FPN barely detect the targets in region #6 of Fig. 8. In contrast, HR-FPN can find out extreme tiny objects further away in scenes with tilted views, as shown in region #1 and #6. In region #2 with tree shading, baseline and FPN detect fewer tiny persons than HR-FPN. Similarly, in region #5, tiny persons only remain a part of their bodies on the water, but this does not hinder the detection of HR-FPN, rather the other two models are severely affected. It indicates that HR-FPN is also robust to feature-deficient targets. Also, the performance of our model is striking in the dense regions that are prone to miss detection, as shown in region #3. In addition, the advantage of HR-FPN is also obvious in the region #4 where the luminance is low. In conclusion, these results demonstrate the strong adaptability of our model on drone-captured scenes.

2) *Quantitative Comparison*: Table VI shows the accuracy of our method compared to other state-of-the-art methods on the test set of VisDrone2019-DET. Related to the medium-sized targets such as “bus” and “car”, “pedestrian” and “people” have smaller scale, so we also report these two categories of targets in Table VI. Obviously, ranking according to the mAP scores, HR-FPN obtains the best results among all models in the Table and surpasses the second place

VarifocalNet by 1.0% on mAP , Qiao *et al.* [52] by 2.0% on mAP^l , and YOLOv5x by 2.4% on mAP^s , which demonstrates the superiority of our method for small-scale object detection. Analyzing the results in Table VI, it can be found that although HR-FPN does not have the best score on mAP^m and mAP^l , it still takes the crown on the overall metric mAP with the excellent small-scale object detection performance, which implies that accurate detection of small-scale objects is critical in the drone-captured scenario. Note that HR-FPN is not optimized specifically for large objects, but its mAP^m , mAP^l scores are not far from their first places, thanks to MSDH.

Table VII shows the results of our model compared with some previous advanced methods on the test set of TinyPerson. To evaluate the small-scale object detection ability of each model in more detail, we also report the results on AP^{t1} , AP^{t2} , AP^{s1} , and AP^{s2} metrics. It can be observed that HR-FPN achieves the highest scores on almost all metrics, especially 1.2% higher than the second place Double Heads on AP , 1.3% ahead of Cascade R-CNN [16] on AP^{t1} , and surpasses TPH-YOLOv5 [17] by 2.2% and 2.5% on AP^{t2} and AP^t , surpasses Grid R-CNN by 1.8% on AP^s . Although the detection results of HR-FPN for medium and large-sized objects are not outstanding, the overall score is more pivotal for the whole scene. And we should take more attention on small-scale objects in aerial scenarios because it is much more difficult to improve the detection accuracy of small-scale objects that lack detailed information than large objects. These

TABLE VII
QUANTITATIVE COMPARISON RESULTS OF PREVIOUS ADVANCED DETECTORS WITH OUR METHOD ON TINYPERSON TEST SET.
BOLD FONT INDICATES THE BEST RESULT OF ONE METRIC

Methods	Backbone	Train Scale	AP^{t1}	AP^{t2}	AP^t	AP^{s1}	AP^{s2}	AP^s	AP^m	AP^l	AP
Two Stage Detectors:											
TridentNet [65]	ResNet-50	Multi	11.7	21.6	19.5	39.7	48.1	44.5	64.8	85.6	41.4
Libra R-CNN [50]	ResNeXt-101	Single	19.1	39.7	33.2	49.2	53.4	52.3	68.6	82.5	44.7
Faster R-CNN (FPN) [19]	ResNet-101	Single	20.0	38.8	33.6	51.5	55.9	52.9	67.1	83.3	45.1
Grid R-CNN [66]	ResNeXt-101	Single	19.9	32.5	29.4	49.9	63.2	59.4	62.1	86.2	46.6
Cascade RPN [67]	ResNet-50	Single	20.5	40.1	34.2	46.6	59.4	52.1	62.3	80.5	47.1
DCN [68]	ResNet-101	Single	23.7	41.7	35.0	53.1	57.9	55.2	68.3	81.6	47.2
Dynamic R-CNN [49]	ResNet-50	Single	23.8	42.6	36.0	53.3	58.8	55.4	66.8	83.8	47.8
CARAFE [69]	ResNet-50	Single	23.2	42.4	35.7	52.6	59.7	55.1	68.4	84.5	47.8
Zhu <i>et al.</i> [70]	ResNet-50	Single	25.9	44.3	37.6	54.2	58.6	55.8	67.1	82.8	48.5
Cascade R-CNN [16]	Res2Net-101 [71]	Single	27.1	41.3	38.6	47.1	60.8	55.9	67.9	86.2	49.0
Cascade R-CNN [16]	ResNeXt-101	Single	26.3	44.8	38.3	52.9	61.5	56.1	68.5	86.7	50.6
Double Heads [39]	ResNet-50	Single	25.5	45.3	39.1	55.3	58.2	56.8	72.3	85.7	51.3
One Stage Detectors:											
RetinaNet [10]	ResNeXt-101	Single	6.2	18.5	14.3	28.2	42.8	33.4	60.3	82.3	28.7
RetinaNet [10]	PVTv2-B4 [72]	Single	5.6	18.8	13.8	34.9	51.0	40.3	66.9	87.8	29.2
YOLOF [20]	ResNet-50	Single	9.2	24.8	21.0	30.5	34.4	30.8	53.6	84.0	30.6
SSD [73]	VGGNet	Single	6.3	22.0	18.2	37.3	46.9	40.8	59.2	64.6	31.0
Deformable DETR [61]	ResNet-50	Multi	11.1	27.7	22.2	39.1	44.0	40.8	56.5	82.0	31.5
SABL [74]	ResNet-101	Multi	6.8	24.4	19.0	39.2	49.0	42.7	62.8	83.6	32.7
RTMDet [75]	CSPNeXt [75]	Single	9.6	25.0	18.4	40.8	49.7	44.4	58.3	76.2	32.9
CenterNet [55]	ResNet-18	Single	8.3	31.1	23.9	47.6	53.2	49.6	65.4	80.8	35.2
PAA [76]	ResNet-50	Multi	10.4	29.3	22.4	43.1	50.0	45.5	64.9	81.0	36.5
ATSS [77]	ResNet-101	Single	18.7	32.9	27.4	44.7	51.5	47.2	65.0	82.2	40.7
Dynamic Head [78]	ResNet-50	Single	21.9	35.6	30.1	47.0	52.9	49.1	68.8	86.1	43.3
FCOS [40]	ResNeXt-101	Multi	20.9	36.4	31.2	47.5	53.2	49.6	65.7	81.8	44.1
RepPoints [79]	ResNeXt-101	Single	16.9	37.2	30.0	51.9	58.1	53.4	68.1	85.8	44.2
DDOD [80]	ResNet-50	Single	25.5	39.1	33.2	52.5	58.6	54.1	67.8	84.7	46.0
LD [81]	ResNet-101	Multi	26.6	42.0	37.3	52.7	58.2	54.4	67.1	76.5	48.5
YOLOv5x	CSP-DarkNet	Single	26.4	40.8	36.9	53.0	57.3	54.4	69.8	85.1	49.2
GFL [82]	ResNeXt-101	Multi	26.5	43.0	38.4	56.5	63.1	58.2	71.5	83.4	50.3
TPH-YOLOv5 [17]	CSP-DarkNet	Single	26.1	47.9	41.4	57.5	61.8	58.6	69.7	76.6	50.4
FSAF [41]	ResNeXt-101	Multi	26.8	42.6	37.7	55.6	61.1	57.8	70.8	88.1	50.8
VarifocalNet [64]	ResNet-101	Multi	26.4	46.1	40.7	57.2	62.6	58.8	71.5	88.2	51.2
HR-FPN (Ours)	CSP-DarkNet	Single	28.4	50.1	43.9	59.6	64.8	61.2	70.9	75.4	52.5

performances show that our series of optimizations have really improved the small-scale object detection.

We also find an interesting phenomenon in the results of Table VI and Table VII that the recently popular ViT models, such as Deformable DETR [61], do not have very prominent performance on the drone-captured scenarios. This is due to the lack of local information in the features as a result of the excessive use of the attention module.

3) *Results Analysis*: The reasons for the excellent results achieved by our approach can be summarized as follows:

- Compared to base FPN, HR-FPN maps all levels of features into high-resolution features with HRFA, which increases the chance of small-scale objects being discovered by shrinking the detection stride.
- HRFF upwardly propagates low-level semantic information for localization, and creates classification and

localization paths to decouple high-resolution features. Compared with multi-scale output, this approach is more suitable for the scenes with mainly small-scale objects.

- The MSDH specially designed for HR-FPN solves the problem of large differences in anchor scales due to single-scale output, and the scale weighted loss function also improves the training effect.

E. Efficiency of Our Methods

To evaluate the efficiency of the proposed method, based on the configurations in section IV-B, we compare of HR-FPN with several of the best-performing models in Table VI and VII on the parameters, GFLOPs, and FPS metrics. The batch size in the test phase is set to 1 to fairly evaluate the FPS. As can be seen from Table VIII, although HR-FPN uses high-resolution features, it has the lowest parameters among all

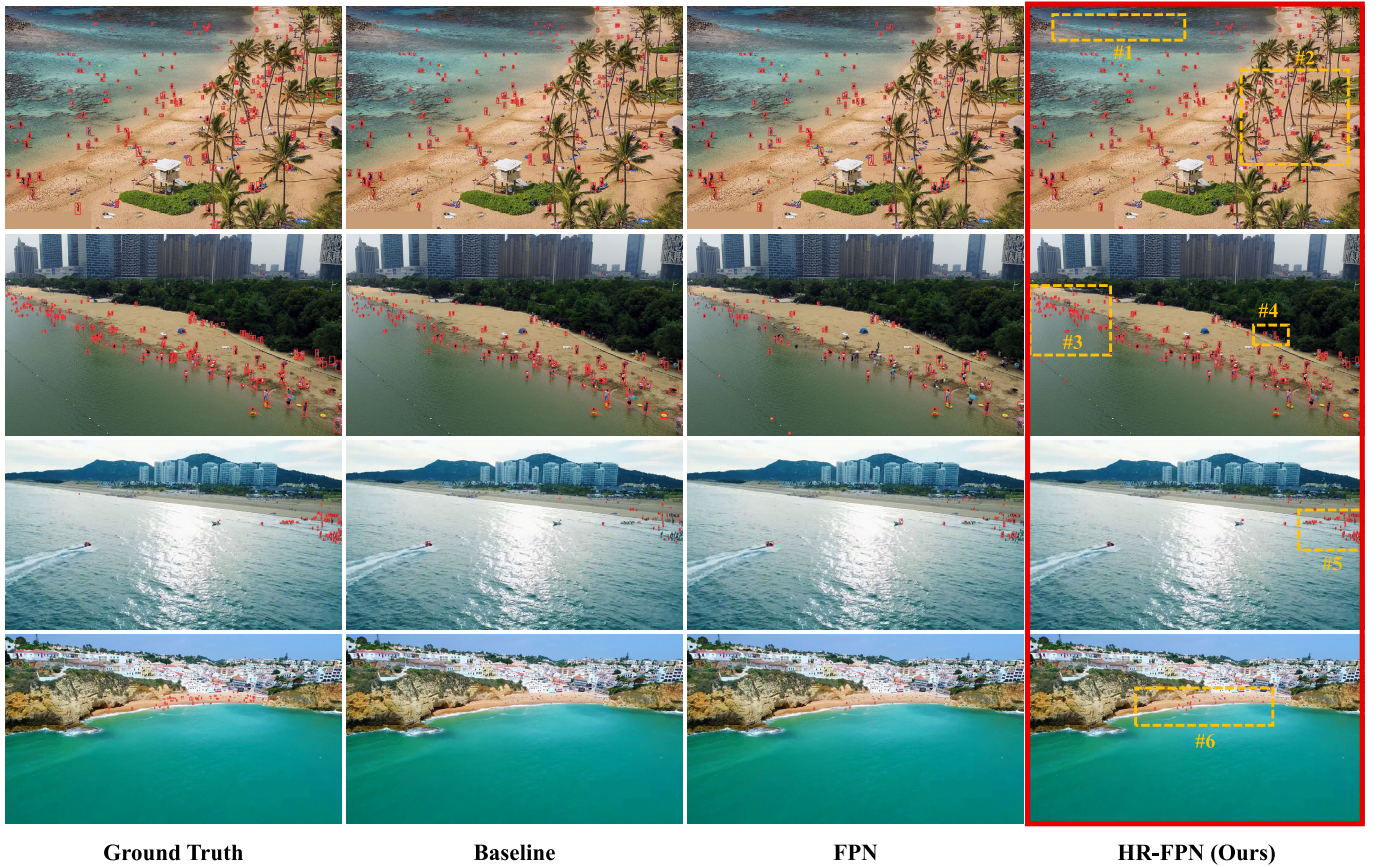


Fig. 8. The qualitative comparison results of different methods on the test set of TinyPerson dataset. For each testing image, we show the results of ground truth, baseline (YOLOv5l), FPN, and our HR-FPN. The orange dotted boxes show our superior performance for tiny object detection in crowded region. Despite containing only one category, the objects in TinyPerson are more difficult to detect due to the extremely small scale.

TABLE VIII
EFFICIENCY COMPARISON RESULTS OF HR-FPN WITH SOME OTHER EXCELLENT MODELS. THE SIZE OF INPUT IMAGE IS 1024×1024 . FPS IS RECORDED WITH GPU ACCELERATION. THE LOWER THE PARAMETERS AND GLOPs THE BETTER, THE HIGHER THE FPS THE BETTER

Methods	Backbone	Params	GFLOPs	FPS
Cascade R-CNN	ResNeXt-101	126.7 M	477.9	12.3
Double Head	ResNet-50	46.7 M	458.5	16.4
TPH-YOLOv5	CSP-DarkNet	93.2 M	650.1	16.2
FSAF	ResNeXt-101	93.7 M	446.1	15.1
VarifocalNet	ResNeXt-101	98.1 M	424.3	11.7
GRoIE	ResNet-50	43.1 M	539.9	12.2
Cascade R-CNN	HRNetV2p-W40	91.0 M	400.1	13.8
YOLOv5l	CSP-DarkNet	46.1 M	276.3	32.8
YOLOv5x	CSP-DarkNet	86.2 M	521.6	23.5
HR-FPN (Ours)	CSP-DarkNet	32.1 M	404.0	23.9

models in the table. This is because we build the HR-FPN network with less convolution layers and channels instead of stacking modules like Cascade R-CNN and TPH-YOLOv5. Compared to baseline (YOLOv5l), our HR-FPN has higher GFLOPs due to the more complex structure. However, in terms of parameters, HR-FPN reduces 30.4% compared to baseline. It is worth noting that the FPS of HR-FPN is about the same as YOLOv5x, but in terms of parameters, HR-FPN is only 37.2% of YOLOv5x, thanks to the lightweight modules we use. These results show that our design achieves a balance

of high performance and efficiency. The lower parameters mean that HR-FPN can be more easily deployed to some edge devices.

V. CONCLUSION AND FUTURE WORK

In this paper, we propose a High-Resolution Feature Pyramid Network (HR-FPN) to improve the accuracy and efficiency of object detection in drone-captured scenarios. We upsample the multi-scale features from backbone into high-resolution features to increase the chance to find out small-scale objects. In addition, the features corresponding to anchors at different scales are used to calculate the locations and categories of objects at corresponding scale. Substantial experimental results show that HR-FPN can effectively improve the detection accuracy of small-scale objects on drone view. However, HR-FPN is insufficient in large-scale object detection. In the future work, we try to make up for this defect by model fusion, and extend our methods to small-scale object tracking.

REFERENCES

- [1] B. Kellenberger, D. Marcos, and D. Tuia, "Detecting mammals in UAV images: Best practices to address a substantially imbalanced dataset with deep learning," *Remote Sens. Environ.*, vol. 216, pp. 139–153, Oct. 2018.
- [2] X. Li, D. K. Giles, F. J. Niederholzer, J. T. Andaloro, E. B. Lang, and L. J. Watson, "Evaluation of an unmanned aerial vehicle as a new method of pesticide application for almond crop protection," *Pest Manage. Sci.*, vol. 77, no. 1, pp. 527–537, Jan. 2021.

- [3] H. Hildmann and E. Kovacs, "Review: Using unmanned aerial vehicles (UAVs) as mobile sensing platforms (MSPs) for disaster response, civil security and public safety," *Drones*, vol. 3, no. 3, p. 59, Jul. 2019.
- [4] E. V. Butilă and R. G. Boboc, "Urban traffic monitoring and analysis using unmanned aerial vehicles (UAVs): A systematic literature review," *Remote Sens.*, vol. 14, no. 3, p. 620, Jan. 2022.
- [5] Z. Zhu, H. Ji, and L. Li, "Deep multi-modal subspace interactive mutual network for specific emitter identification," *IEEE Trans. Aerosp. Electron. Syst.*, early access, Jan. 26, 2023, doi: 10.1109/TAES.2023.3240115.
- [6] Y. Cao, H. Ji, W. Zhang, and S. Shirani, "Feature aggregation networks based on dual attention capsules for visual object tracking," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 2, pp. 674–689, Feb. 2022.
- [7] Z. Zhu, H. Ji, W. Zhang, L. Li, and T. Ji, "Complex convolutional neural network for signal representation and its application to radar emitter recognition," *IEEE Commun. Lett.*, vol. 27, no. 3, pp. 856–860, Mar. 2023.
- [8] L. Li, Z. Dong, Z. Zhu, and Q. Jiang, "Deep-learning hopping capture model for automatic modulation classification of wireless communication signals," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 59, no. 2, pp. 772–783, Apr. 2023.
- [9] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 28, 2015, pp. 1–9.
- [10] T. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.
- [11] J. Leng, M. Mo, Y. Zhou, C. Gao, W. Li, and X. Gao, "Pareto refocusing for drone-view object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 3, pp. 1320–1334, Mar. 2023.
- [12] X. Yu, Y. Gong, N. Jiang, Q. Ye, and Z. Han, "Scale match for tiny person detection," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Mar. 2020, pp. 1257–1265.
- [13] P. Zhu et al., "Detection and tracking meet drones challenge," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 11, pp. 7380–7399, Nov. 2022.
- [14] T.-Y. Lin et al., "Microsoft COCO: Common objects in context," in *Proc. Eur. Conf. Comput. Vis. Zürich, Switzerland: Springer*, 2014, pp. 740–755.
- [15] M. Everingham, L. Van Gool, C. K. I. Williams, J. Winn, and A. Zisserman, "The PASCAL visual object classes (VOC) challenge," *Int. J. Comput. Vis.*, vol. 88, no. 2, pp. 303–338, Jun. 2010.
- [16] Z. Cai and N. Vasconcelos, "Cascade R-CNN: High quality object detection and instance segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 5, pp. 1483–1498, May 2021.
- [17] X. Zhu, S. Lyu, X. Wang, and Q. Zhao, "TPH-YOLOv5: Improved YOLOv5 based on transformer prediction head for object detection on drone-captured scenarios," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2021, pp. 2778–2788.
- [18] K. Sun, B. Xiao, D. Liu, and J. Wang, "Deep high-resolution representation learning for human pose estimation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 5693–5703.
- [19] T. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2117–2125.
- [20] Q. Chen, Y. Wang, T. Yang, X. Zhang, J. Cheng, and J. Sun, "You only look one-level feature," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 13039–13048.
- [21] C. Wang, H. M. Liao, Y. Wu, P. Chen, J. Hsieh, and I. Yeh, "CSPNet: A new backbone that can enhance learning capability of CNN," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2020, pp. 390–391.
- [22] S. Liu, L. Qi, H. Qin, J. Shi, and J. Jia, "Path aggregation network for instance segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8759–8768.
- [23] Z. Li, C. Lang, J. H. Liew, Y. Li, Q. Hou, and J. Feng, "Cross-layer feature pyramid network for salient object detection," *IEEE Trans. Image Process.*, vol. 30, pp. 4587–4598, 2021.
- [24] C. Deng, M. Wang, L. Liu, Y. Liu, and Y. Jiang, "Extended feature pyramid network for small object detection," *IEEE Trans. Multimedia*, vol. 24, pp. 1968–1979, 2022.
- [25] J. Leng, Y. Ren, W. Jiang, X. Sun, and Y. Wang, "Realize your surroundings: Exploiting context information for small object detection," *Neurocomputing*, vol. 433, pp. 287–299, Apr. 2021.
- [26] S. Huang and Q. Liu, "Addressing scale imbalance for small object detection with dense detector," *Neurocomputing*, vol. 473, pp. 68–78, Feb. 2022.
- [27] C. Yang, Z. Huang, and N. Wang, "QueryDet: Cascaded sparse query for accelerating high-resolution small object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 13668–13677.
- [28] X. Liang, J. Zhang, L. Zhuo, Y. Li, and Q. Tian, "Small object detection in unmanned aerial vehicle images using feature fusion and scaling-based single shot detector with spatial context analysis," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 30, no. 6, pp. 1758–1770, Jun. 2020.
- [29] K. Duan, D. Du, H. Qi, and Q. Huang, "Detecting small objects using a channel-aware deconvolutional network," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 30, no. 6, pp. 1639–1652, Jun. 2020.
- [30] A. Oliva and A. Torralba, "The role of context in object recognition," *Trends Cogn. Sci.*, vol. 11, no. 12, pp. 520–527, Dec. 2007.
- [31] X. Tang, D. K. Du, Z. He, and J. Liu, "PyramidBox: A context-assisted single shot face detector," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 797–813.
- [32] J.-S. Lim, M. Astrid, H.-J. Yoon, and S.-I. Lee, "Small object detection using context and attention," in *Proc. Int. Conf. Artif. Intell. Inf. Commun. (ICAIIIC)*, Apr. 2021, pp. 181–186.
- [33] Q. Yin, W. Yang, M. Ran, and S. Wang, "FD-SSD: An improved SSD object detection algorithm based on feature fusion and dilated convolution," *Signal Process., Image Commun.*, vol. 98, Oct. 2021, Art. no. 116402.
- [34] J. Nie, Y. Pang, S. Zhao, J. Han, and X. Li, "Efficient selective context network for accurate object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 31, no. 9, pp. 3456–3468, Sep. 2021.
- [35] Z. Xia, X. Pan, S. Song, L. E. Li, and G. Huang, "Vision transformer with deformable attention," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 4794–4803.
- [36] X. Ding, X. Zhang, J. Han, and G. Ding, "Scaling up your kernels to 31×31 : Revisiting large kernel design in CNNs," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 11963–11975.
- [37] X. Pan et al., "On the integration of self-attention and convolution," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 815–825.
- [38] G. Song, Y. Liu, and X. Wang, "Revisiting the sibling head in object detector," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 11563–11572.
- [39] Y. Wu et al., "Rethinking classification and localization for object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10186–10195.
- [40] Z. Tian, C. Shen, H. Chen, and T. He, "FCOS: Fully convolutional one-stage object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9627–9636.
- [41] C. Zhu, Y. He, and M. Savvides, "Feature selective anchor-free module for single-shot object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 840–849.
- [42] X. Yang, J. Yan, W. Liao, X. Yang, J. Tang, and T. He, "SCRDet++: Detecting small, cluttered and rotated objects via instance-level feature denoising and rotation loss smoothing," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 2, pp. 2384–2399, Feb. 2023.
- [43] A. Bochkovskiy, C.-Y. Wang, and H.-Y. M. Liao, "YOLOv4: Optimal speed and accuracy of object detection," 2020, *arXiv:2004.10934*.
- [44] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
- [45] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.
- [46] Z. Zheng, P. Wang, W. Liu, J. Li, R. Ye, and D. Ren, "Distance-IoU loss: Faster and better learning for bounding box regression," in *Proc. AAAI Conf. Artif. Intell.*, Apr. 2020, vol. 34, no. 7, pp. 12993–13000. [Online]. Available: <https://ojs.aaai.org/index.php/AAAI/article/view/6999>
- [47] D. P. Papadopoulos, J. R. R. Uijlings, F. Keller, and V. Ferrari, "We don't need no bounding-boxes: Training object class detectors using only human verification," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 854–863.
- [48] D. Du et al., "The unmanned aerial vehicle benchmark: Object detection and tracking," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Sep. 2018, pp. 370–386.

- [49] H. Zhang, H. Chang, B. Ma, N. Wang, and X. Chen, "Dynamic R-CNN: Towards high quality object detection via dynamic training," in *Proc. Eur. Conf. Comput. Vis.* Glasgow, U.K.: Springer, 2020, pp. 260–275.
- [50] J. Pang, K. Chen, J. Shi, H. Feng, W. Ouyang, and D. Lin, "Libra R-CNN: Towards balanced learning for object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 821–830.
- [51] I. Radosavovic, R. P. Kosaraju, R. Girshick, K. He, and P. Dollár, "Designing network design spaces," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10428–10436.
- [52] S. Qiao, H. Wang, C. Liu, W. Shen, and A. Yuille, "Micro-batch training with batch-channel normalization and weight standardization," 2019, *arXiv:1903.10520*.
- [53] Y. Cao, K. Chen, C. C. Loy, and D. Lin, "Prime sample attention in object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 11583–11591.
- [54] L. Rossi, A. Karimi, and A. Prati, "A novel region of interest extraction layer for instance segmentation," in *Proc. 25th Int. Conf. Pattern Recognit. (ICPR)*, Jan. 2021, pp. 2203–2209.
- [55] X. Zhou, D. Wang, and P. Krähenbühl, "Objects as points," 2019, *arXiv:1904.07850*.
- [56] D. Meng et al., "Conditional DETR for fast training convergence," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 3651–3660.
- [57] B. Zhu et al., "AutoAssign: Differentiable label assignment for dense object detection," 2020, *arXiv:2007.03496*.
- [58] S. Liu et al., "DAB-DETR: Dynamic anchor boxes are better queries for DETR," in *Proc. Int. Conf. Learn. Represent.*, 2022, pp. 1–9. [Online]. Available: <https://openreview.net/forum?id=oMI9PjOb9Jl>
- [59] N. Wang et al., "NAS-FCOS: Fast neural architecture search for object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 11943–11951.
- [60] X. Zhang, F. Wan, C. Liu, R. Ji, and Q. Ye, "FreeAnchor: Learning to match anchors for visual object detection," in *Advances in Neural Information Processing Systems*, vol. 32, H. Wallach, H. Larochelle, A. Beygelzimer, F. d'Alché-Buc, E. Fox, and R. Garnett, Eds. Red Hook, NY, USA: Curran Associates, 2019. [Online]. Available: <https://proceedings.neurips.cc/paper/2019/file/43ec517d68b6edd3015b3edc9a11367b-Paper.pdf>
- [61] X. Zhu, W. Su, L. Lu, B. Li, X. Wang, and J. Dai, "Deformable DETR: Deformable transformers for end-to-end object detection," 2020, *arXiv:2010.04159*.
- [62] C. Feng, Y. Zhong, Y. Gao, M. R. Scott, and W. Huang, "TOOD: Task-aligned one-stage object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 3490–3499.
- [63] H. Zhang et al., "DINO: DETR with improved denoising anchor boxes for end-to-end object detection," 2022, *arXiv:2203.03605*.
- [64] H. Zhang, Y. Wang, F. Dayoub, and N. Sunderhauf, "VarifocalNet: An IoU-aware dense object detector," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 8514–8523.
- [65] Y. Li, Y. Chen, N. Wang, and Z.-X. Zhang, "Scale-aware trident networks for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6054–6063.
- [66] X. Lu, B. Li, Y. Yue, Q. Li, and J. Yan, "Grid R-CNN," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 7363–7372.
- [67] T. Vu, H. Jang, T. X. Pham, and C. Yoo, "Cascade RPN: Delving into high-quality region proposal network with adaptive convolution," in *Advances in Neural Information Processing Systems*, vol. 32, H. Wallach, H. Larochelle, A. Beygelzimer, F. d'Alché-Buc, E. Fox, and R. Garnett, Eds. Red Hook, NY, USA: Curran Associates, 2019. [Online]. Available: <https://proceedings.neurips.cc/paper/2019/file/d554f7bb7be44a7267068a7df88dd20-Paper.pdf>
- [68] J. Dai et al., "Deformable convolutional networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 764–773.
- [69] J. Wang, K. Chen, R. Xu, Z. Liu, C. C. Loy, and D. Lin, "CARAFE: Content-aware aassembly of features," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 1–10.
- [70] X. Zhu, D. Cheng, Z. Zhang, S. Lin, and J. Dai, "An empirical study of spatial attention mechanisms in deep networks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6688–6697.
- [71] S.-H. Gao, M.-M. Cheng, K. Zhao, X.-Y. Zhang, M.-H. Yang, and P. Torr, "Res2Net: A new multi-scale backbone architecture," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 2, pp. 652–662, Feb. 2021.
- [72] W. Wang et al., "PVT v2: Improved baselines with pyramid vision transformer," *Comput. Vis. Media*, vol. 8, no. 3, pp. 415–424, Sep. 2022.
- [73] W. Liu et al., "SSD: Single shot multibox detector," in *Proc. Eur. Conf. Comput. Vis.* Amsterdam, The Netherlands: Springer, 2016, pp. 21–37.
- [74] J. Wang et al., "Side-aware boundary localization for more precise object detection," in *Proc. Eur. Conf. Comput. Vis.* Glasgow, U.K.: Springer, 2020, pp. 403–419.
- [75] C. Lyu et al., "RTMDet: An empirical study of designing real-time object detectors," 2022, *arXiv:2212.07784*.
- [76] K. Kim and H. S. Lee, "Probabilistic anchor assignment with IoU prediction for object detection," in *Proc. Eur. Conf. Comput. Vis.* Glasgow, U.K.: Springer, 2020, pp. 355–371.
- [77] S. Zhang, C. Chi, Y. Yao, Z. Lei, and S. Z. Li, "Bridging the gap between anchor-based and anchor-free detection via adaptive training sample selection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9759–9768.
- [78] X. Dai et al., "Dynamic head: Unifying object detection heads with attentions," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 7373–7382.
- [79] Z. Yang, S. Liu, H. Hu, L. Wang, and S. Lin, "RepPoints: Point set representation for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9657–9666.
- [80] Z. Chen, C. Yang, Q. Li, F. Zhao, Z.-J. Zha, and F. Wu, "Disentangle your dense object detector," in *Proc. 29th ACM Int. Conf. Multimedia*, Oct. 2021, pp. 4939–4948.
- [81] Z. Zheng et al., "Localization distillation for dense object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 9407–9416.
- [82] X. Li et al., "Generalized focal loss: Learning qualified and distributed bounding boxes for dense object detection," in *Advances in Neural Information Processing Systems*, vol. 33, H. Larochelle, M. Ranzato, R. Hadsell, M. Balcan, and H. Lin, Eds. Red Hook, NY, USA: Curran Associates, 2020, pp. 21002–21012. [Online]. Available: <https://proceedings.neurips.cc/paper/2020/file/f0bda020d2470f2e74990a07a607ebd9-Paper.pdf>
- [83] K. Han, Y. Wang, Q. Tian, J. Guo, C. Xu, and C. Xu, "GhostNet: More features from cheap operations," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 1580–1589.



Zhaodong Chen (Student Member, IEEE) received the B.S. degree from the School of Electronic Engineering, Xidian University, Xi'an, China, in 2020, where he is currently pursuing the Ph.D. degree in circuit and system. His current research interests include computer vision, machine learning, object detection, and tracking.



Hongbing Ji (Senior Member, IEEE) was born in Shaanxi, China, in 1963. He received the B.S. degree in radar engineering from the Northern West Telecommunications Engineering College, Xidian University, in 1983, and the M.S. degree in circuit, signals and systems and the Ph.D. degree in signal and information processing from Xidian University in 1989 and 1999, respectively.

After graduation in 1989, he was with the School of Electronic Engineering, Xidian University, a Lecturer, from 1990 to 1995, an Associate Professor from 1995 to 2000, and a Professor since 2000. From 1996 to 2002, he was the Vice Dean of the School of Electronic Engineering. From 2002, he was the Executive Dean of the Graduate School, Xidian University, and the Vice Chairperson of the Academic Degree Evaluation Committee. His research interests include radar signal processing, automatic targets recognition, multi-sensor information fusion, and target tracking.

Prof. Ji is a member of the IEEE Signal Processing Society and a member of the IEEE Aerospace and Electronic Systems Society.



Yongquan Zhang was born in Gansu, China, in 1985. He received the B.S. degree in computer science and technology and the M.S. degree in computer applications technology from the Lanzhou University of Technology in 2007 and 2010, respectively, and the Ph.D. degree in intelligent information processing from Xidian University in 2014. Currently, he is an Associate Professor with the School of Electronic Engineering, Xidian University. His research interests include target tracking, information fusion, machine learning, and signal processing.



Yifan Li received the bachelor's degree from Xidian University, Xi'an, China, in 2016, where he is currently pursuing the Ph.D. degree in intelligent information processing. His current research interests include computer vision, object detection, and multi-object tracking.



Zhigang Zhu (Member, IEEE) was born in Shandong, China, in 1989. He received the B.S. degree from the School of Communication and Electronic Engineering, Qingdao University of Technology, in 2013, the M.S. degree from the School of Telecommunications Engineering, Xidian University, in 2015, and the Ph.D. degree from the School of Electronic Engineering, Xidian University, in 2020. Currently, he is a Lecturer with the School of Electronic Engineering, Xidian University. His research interests include deep learning, pattern

recognition, and signal processing.